Boxuan Wang

Design Engineering Student

Robotics | Intelligent Systems | jw4424@ic.ac.uk | mystify59.github.io

PROFILE

Design Engineering student at Imperial College London with experience across product design, humanoid development, robot testing and Python. I combine creative problem-solving with technical analysis, CAD and hands-on prototyping to develop practical solutions that address both user needs and engineering challenges.

EDUCATION

Imperial College London 2024 - 2028

Design Engineering | Expected completion 2028

Relevant study: Intelligent Systems, Physical Computing, Mechanical Engineering, FEA.

The King's School, Canterbury 2019 - 2024

King's Scholar; Vice Captain of Fencing.

WORK EXPERIENCE

Li Auto, *Humanoid Development Intern* July - September 2026

- Designed CCU, microphone and other component supports in SolidWorks, resolving fixing positions, clearances and assembly access.
- Investigated main-frame load paths in Ansys, comparing static stress, deformation and local mesh sensitivity under stated load assumptions.
- Designed a wrist-camera mount and analysed static load cases; supported VLA demonstration capture using wearable tracking, VR and robot arms.

Elibot, *Cooperative Robot Intern* July - August 2023

- Created joint-specific fatigue tests, logged faults and examined angular differences during braking in harmonic-drive joints.
- Built a Python system to log incoming, produced and shipped robots.
- Contributed to app and web interfaces for remote robot control and webpage UX.

SELECTED PROJECTS

Autonomous Football Robot, *Team robotics project* 2025

- Contributed to team model training and ball-tracking code using OpenMV and Edge Impulse.
- Contributed to team control-code development for autonomous navigation and task execution on an existing robot platform.

Street Pulse, *Interactive floor for street performance* 2025

- Carried out almost all user research: interviews, observations and audience heat maps; developed the tessellation and exterior design.
- Prototyped the electronics, assembled the functional vibration module and carried out physical testing; other team members developed the software.
- Tested the module with 19 participants alongside the team's 3D street-performance video viewed in VR to assess the proposed in-situ experience.

Thermofluids, *Vehicle & battery cooling* 2026

- Researched SUV aerodynamics, developed profile alternatives and built simplified Fusion geometry for an initial Fluent study of rear-wake behaviour.
- Researched channel aspect ratio and iterated the team's cooling-plate CAD; assisted with CFD and interpreted pressure-loss and temperature trade-offs.

SKILLS

Programming and vision. Python, C++, MATLAB; OpenMV, Edge Impulse.

Mechanical design. SolidWorks, Fusion; CAD, design for additive manufacturing, prototype assembly.

Analysis and testing. Ansys Mechanical, Fluent; FEA, CFD and robot-joint fatigue testing.

AI tools. AI agents and MCP tools for 3D-printable validation structures and temporary prototypes.

LANGUAGES

Chinese (Native) | **English** (Fluent) | **French** (Basic)